Title	Authors	Year	Dataset	Application Area	Method/Model	Metrics (Result in numbers)	Strengths	Limitations	Research Gap
"Semi-Supervised Disease Classification based on Limited Medical Image Data"	Yan Zhang, Chun Li, Zhaoxia Liu, Ming Li.	2024.	The datasets used are BreastMNIST, PneumoniaMNIST, BloodMNIST, OCTMNIST, and AMD..	The application area is medical image classification and semi-supervised disease classification..	The model introduced is the HD-PAN (Hölder Divergence Predictive Adversarial Networks)..	On BreastMNIST, it achieved 0.8141 accuracy and 0.8755 F1-score.. On PneumoniaMNIST, it reached 0.8782 accuracy and 0.9003 F1-score..	It outperforms existing KL divergence approaches, does not rely on prior probabilities, and effectively leverages limited labeled data..	The model still relies on a simple 0-1 binary classification and may face challenges depending on diverse dataset characteristics..	Future directions include integrating diffusion models for expanding positive samples, exploring new objective functions, and addressing class imbalances..
"Dynamic Filter Application in Graph Convolutional Networks for Enhanced Spectral Feature Analysis and Class Discrimination in Medical Imaging"	Aryan Singh, Pepijn van de Ven, Ciarán Eising, Patrick Denny.	2024.	The study utilizes the MedMNIST collection, MNIST, CIFAR-10, and MSTAR-10 datasets..	The application is medical image analysis and graph-level classification..	The proposed method is GCNN-EC (Graph Convolution Neural Network Enhanced Connectivity) with a dynamic filter layer..	It achieves competitive accuracy (e.g., BreastMNIST accuracy of 0.795) while using up to 100 times fewer parameters than traditional CNNs (95,210 vs 11,689,512)..	It successfully mitigates the over-smoothing problem common in GCNs and maintains high accuracy with a computationally efficient structure..	The model is currently constrained by simple node representations relying strictly on individual pixels..	Further research is needed to explore sophisticated node representations (like bounding boxes) and integrate the dynamic filter layer with Vision Transformers (ViTs)..
"Epidemic Forecasting via Hybrid Deep Learning With Unified Visibility and Temporal	Arman Kavoosi Ghafi, Ali Pirkhedri, Samira Akhbarifar, Mohammad	2026.	It uses the Johns Hopkins University COVID-19 global confirmed case time series..	The focus is on epidemic forecasting and public health monitoring..	The framework uses a Temporal Graph Network with Visibility Graphs (TGN-VG) fused with	The model achieved an MAE of 0.0558, RMSE of 0.0709, MAPE of 73.97%, and R^2 of 0.83 on clean 7-day	It is highly robust against data corruption, missing data, and noise, while a cloud-native deployment	The evaluation focuses primarily on aggregate global trends, which may overlook unique patterns found at	Future work should explore real-time multi-region deployment, the integration of mobility or

Graphs Under Stochastic Noise Modelling"	Hossein Shafiabadi.				stochastic noise modelling..	forecasting horizons..	reduces inference time by over 60% and memory usage by 35%..	the individual or regional levels..	vaccination data, and the optimization of edge-cloud hybrid architectures..
"Color-Guided Mixture-of-Experts Conditional GAN for Realistic Biomedical Image Synthesis in Data-Scarce Diagnostics"	Patrycja Kwiek, Filip Ciepiela, Małgorzata Jakubowska.	2025.	The research utilizes RGB microscopic images from the BloodMNIST dataset at a 3 × 128 × 128 resolution..	The area is biomedical image synthesis and data augmentation for hematological diagnostics..	The study proposes a Color-Guided Mixture-of-Experts Conditional GAN (MoE-cGAN) architecture incorporating a Wasserstein histogram loss..	Synthetic images achieved a classification accuracy of 0.97, precision of 0.96, recall of 0.97, F1-score of 0.96, and a best Fréchet Inception Distance (FID) of 52.1..	The modular expert specialization mitigates mode collapse, while targeted red and green color histogram alignment significantly improves the realism of synthetic images..	Misclassifications still occur among visually and chromatically similar cell types, and the inclusion of the blue channel was found to introduce unwanted noise..	Potential improvements include integrating spatial attention mechanisms, adaptive color normalization for different staining protocols, and multimodal feature conditioning..
"IncARMAG: A convolutional neural network with multi-level autoregressive moving average graph convolutional processing framework for medical image classification"	Adrian S. Remigio.	2024.	The model was evaluated on 2D MedMNIST datasets (DermaMNIST, OCTMNIST, PneumoniaMNIST, BreastMNIST, and BloodMNIST)..	The primary application is medical image classification across various imaging modalities..	The proposed IncARMAG integrates an Inception V3 backbone, a Self-Constructing Graph (SCG) module, and Autoregressive Moving Average (ARMA) graph convolutions..	The model achieved top accuracies such as 0.802 on DermaMNIST, 0.873 on OCTMNIST, 0.885 on PneumoniaMNIST, 0.872 on BreastMNIST, and 0.987 on BloodMNIST..	It outperforms various state-of-the-art CNN, Transformer, and hybrid models by effectively extracting multi-level hierarchical features and leveraging long-range dependencies..	The current study lacks data augmentation and hyperparameter optimization, and requires substantial prospective clinical validation..	Future directions point to employing data augmentation, hyperparameter optimization, addressing algorithmic biases, and applying the architecture as a backbone for image segmentation..

[1]. Zhang, Y., Li, C., Liu, Z., & Li, M. (2024). Semi-supervised disease classification based on limited medical image data. IEEE Journal of Biomedical and Health Informatics, 28, 1575–1586.
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[3]. Ghafi, A. K., Pirkhedri, A., Akhbarifar, S., & Shafiabadi, M. H. (2026). Epidemic forecasting via hybrid deep learning with unified visibility and temporal graphs under stochastic noise modelling. Healthcare Technology Letters.
[4]. Kwiek, P., Ciepiela, F., & Jakubowska, M. (2025). Color-guided mixture-of-experts conditional GAN for realistic biomedical image synthesis in data-scarce diagnostics. Electronics, 14, 2773. https://doi.org/10.3390/electronics14142773
[5]. Remigio, A. S. (2024). IncARMAG: A convolutional neural network with multi-level autoregressive moving average graph convolutional processing framework for medical image classification [Preprint submitted to Computers in Biology and Medicine].